CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 4: **Quadratic Equations** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If one root of $x^2 - 11x + k = 0$ is 5, then k is: 1

(a) 11 **(b)** 30 **(c)** 6 **(d)** 35

2. The roots of $x^2 + 2x - 15 = 0$ are: 1

(a) -5 and 3 **(b)** 5 and -3 **(c)** 5 and 3 **(d)** -5 and -3

3. The discriminant of $4x^2 + 4x + 1 = 0$ is: 1

(a) 12 **(b)** 32 **(c)** 0 **(d)** 8

4. The nature of the roots of $4x^2 - 4x + 1 = 0$ is: 1

(a) two distinct real roots **(b)** no real roots

(c) three real roots **(d)** two equal real roots

5. For the equation $x^2 - 13x + 36 = 0$, the sum of the roots is: 1

(a) 36 **(b)** -13 **(c)** -23 **(d)** 13

6. For $x^2 - 14x + k = 0$ to have equal roots, $k =$ 1

(a) 7 **(b)** 49 **(c)** 14 **(d)** 196

7. If one root of $x^2 - 16x + k = 0$ is 6, then k is: 1

(a) 60 **(b)** 66 **(c)** 10 **(d)** 16

8. The roots of $x^2 - 11x + 10 = 0$ are: 1

(a) 1 and 10 **(b)** -1 and 10 **(c)** -1 and -10 **(d)** 1 and -10

9. The discriminant of $5x^2 + 3x - 1 = 0$ is: 1

(a) 19 **(b)** -11 **(c)** -29 **(d)** 29

10. The nature of the roots of $2x^2 + 5x + 4 = 0$ is: 1

(a) three real roots **(b)** two equal real roots

(c) no real roots **(d)** two distinct real roots

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find k for which $x^2 - 12x + k = 0$ has two equal roots. 2

12. Represent the situation as a quadratic equation: The product of two consecutive positive integers is 156. 2

13. Find the roots of $2x^2 + 3x - 5 = 0$ by factorisation. 2

14. Find k for which $x^2 - 16x + k = 0$ has two equal roots. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the roots of $x^2 + 2x + 5 = 0$ using the quadratic formula (if they exist). 3

16. Find the roots of $x^2 + 6x - 7 = 0$ using the quadratic formula (if they exist). 3

17. Find the values of k for which $x^2 + kx + 36 = 0$ has equal roots. 3

OR

Find the roots of $5x^2 - 3x - 2 = 0$ using the quadratic formula (if they exist). 3

18. Find the roots of $3x^2 - 5x - 2 = 0$ using the quadratic formula (if they exist). 3

OR

Find the values of k for which $x^2 + kx + 49 = 0$ has equal roots. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The altitude of a right triangle is 4 cm less than its base and the hypotenuse is 20 cm. Find the base and altitude. 5

OR

The sum of the ages of two friends is 36 years. 6 years ago the product of their ages was 63. Find their present ages. 5

20. The altitude of a right triangle is 3 cm less than its base and the hypotenuse is 15 cm. Find the base and altitude. 5

OR

The sum of the ages of two friends is 49 years. 7 years ago the product of their ages was 196. Find their present ages. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND